



X-Raying the ancients

How handheld x-ray fluorescence machines help archaeologists explore ancient wall paintings. <https://digit.in/jun20-64>



The past we can never return to

A look at the meaning behind one of the oldest and simplest of artistic practices, leaving handprints. <https://digit.in/jun20-65>

Technology for Archaeology

Using cutting edge tools to explore the past

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rchaeologists have to do all the things that scientists do in the present time, but for the past. As a

consequence, it is a very interdisciplinary field, and archaeologists have become very good at using tools and technologies developed for other applications. The approach now is to get as much information from a find as possible without destroying it, and studying whatever is possible where it is found. Such investigations are made possible because of miniaturisation of equipment. New technologies are also allowing archaeologists to get new information on finds that are already well-known and studied. Here are some of the newest technologies that archaeologists are using.

PHASE-CONTRAST X-RAY IMAGING

Traditional X-ray imaging techniques use a machine to pass X-rays through an object. Some materials such as bones absorb more X-rays while flesh absorbs less. A relatively less common aspect of X-rays is that the stream of particles change direction as they pass through an object of study, similar to how light changes direction when passed through a lens. Like the absorption contrast slides, this property of X-rays can also provide phase-contrast slides.



A Herculaneum scroll.

The imaging technique is used to study one of the most interesting types of archaeological finds – mummies. When it comes to mummies, Phase-contrast X-ray imaging is used in tandem with CT scans that can explore soft tissues very well. The approach was used to scan the hand of an Egyptian mummy, down to a resolution of 6 to 9 micrometers, which is a little more than the size of a red blood cell. The embalming process was so successful that the nerves and blood vessels within the hand were also preserved. With a cellular level resolution, researchers can potentially find signs of diseases in other mummies. The Egyptians mummified all sorts of animals and the approach was used to study a rare mummy of a Crocodile housed at the Louvre. The study showed that the Egyptians had removed most of the internal organs of the crocodile, and even most of the skeleton, and had replaced it with plant material.

MULTISPECTRAL IMAGING

The human eye can only see things in a limited portion of the electromagnetic spectrum. Multispectral imaging goes beyond what the human eye can see, and combines information from infrared and ultraviolet portions of the spectrum as well. Initially, the technology was developed by NASA to study mineral deposits on the Earth, and later on other planets. The technology has since become cheaper, more accessible, and smaller as well. In Archeology, combining infrared, X-ray and visible light information in a single system allows researchers to decipher ancient documents in ways that were never before possible. This is because the really old documents are found in all kinds of ridiculous states. They may be the dried remains of a soggy pulp, the ink might have faded, the pages may be stuck together, the documents might be little more than charred remains, or they may be tightly wound

Credit: Seth Parker/ University of Kentucky